

Model for Calculating the Edelstein Effect in a Rashba Fermion System

1. Theoretical Framework and Hamiltonian

The system is modeled as a two-dimensional electron gas (2DEG) with Rashba spin-orbit coupling (RSOC). The Hamiltonian describing the Rashba fermion at the Γ point of the Brillouin zone is given by:

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma}) \quad (1)$$

where:

- p is the momentum operator.
- m is the effective carrier mass.
- α is the Rashba spin-orbit coupling strength.
- $\hat{z}$ is the unit vector perpendicular to the 2D plane.
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the vector of Pauli matrices.

The energy dispersion relation for the two chiral (helicity) bands $\nu = \pm$ is:

$$E_\nu(k) = \frac{\hbar^2 k^2}{2m} + \nu \alpha k \quad (6)$$

where $k = |\mathbf{k}|$ and $\nu = +1$ corresponds to the outer branch and $\nu = -1$ to the inner branch (or vice versa depending on the sign convention of α). The helicity operator is defined as $\hat{S} = \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})/p$, with eigenvalues $s = \pm 1$ [1].

2. Calculation of Magnetization (Edelstein Effect)

The Direct Edelstein Effect (DEE) describes the generation of a non-equilibrium spin density (magnetization) $\mathbf{M}$ under an applied electric field $\mathbf{E}$. Within the semiclassical Boltzmann approach, the magnetization at first order in the electric field is given by [1]:

$$\mathbf{M} = -\mu_b \sum_{\mathbf{k}, \nu} |e| (\mathbf{v}_\nu(\mathbf{k}) \cdot \mathbf{E}) \delta[E_\nu(\mathbf{k}) - E_F] \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^\nu \quad (2)$$

where:

- μ_b is the Bohr magneton.
- $|e|$ is the elementary charge.
- $\mathbf{v}_\nu(\mathbf{k}) = \nabla_{\mathbf{k}} E_\nu(\mathbf{k})$ is the group velocity.
- $\bar{\tau}_{\mathbf{k}}^\nu$ is the transport lifetime (often assumed constant τ for simplicity).
- $\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^\nu$ is the spin expectation value for the eigenstates.

For the isotropic Rashba model, the spin expectation value is:

$$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^\pm = \frac{1}{k} \begin{pmatrix} \pm k_y \\ \mp k_x \\ 0 \end{pmatrix} = \begin{pmatrix} \pm \sin \theta \\ \mp \cos \theta \\ 0 \end{pmatrix} \quad (3)$$

where θ is the angle between $\mathbf{k}$ and the $\hat{x}$ axis.

2.1. Magnetization Magnitude and Direction

Assuming an electric field applied along the $\hat{x}$ direction, $\mathbf{E} = E_x \hat{x}$, the resulting magnetization is perpendicular to the electric field and lies in the plane (along $\hat{y}$). The magnitude depends on the filling regime:

High-Density Regime (HDR)

When both chiral bands are occupied ($E_F > 0$ relative to the band crossing):

$$M_y = \frac{\mu_b |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y \quad (8)$$

In this regime, the spin density is **constant** and independent of the Fermi energy E_F . It scales linearly with the spin-orbit coupling α and the effective mass m .

Low-Density Regime (LDR)

When only the lowest energy band is occupied ($E_F < 0$ relative to the band crossing, or near the band minimum):

$$M_y = \frac{\mu_b |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y \quad (9)$$

In this regime, the spin density increases linearly with the Fermi energy E_F for small E_F . For values of E_F around the band crossing, the expression can be expanded as:

$$M_y \approx \frac{\mu_b |e| \tau}{2\pi} \left(\alpha m + \frac{1}{2} \frac{E_F}{\alpha} \right) [\hat{z} \times \mathbf{E}]_y \quad (10)$$

2.2. Edelstein Susceptibility

The linear Edelstein susceptibility χ_{ij} is defined by $M_j = \chi_{ij} E_i$. For the isotropic case with $\mathbf{E} = E_x \hat{x}$:

$$\chi_{xy} = -\chi_0 \sum_{\nu=\pm} \int d^2 k \langle \sigma_y \rangle_{\mathbf{k}}^{\nu} \delta(E_{\mathbf{k}}^{\nu} - \mu) v_x^{\nu}(\mathbf{k}) \quad (7)$$

where $\chi_0 = \frac{\tau |e| \mu_b S_{cell}}{4\pi^2 a}$.

Parameter Dependencies:

- **Spin-Orbit Coupling (α):** In the HDR, χ_{xy} increases linearly with α (see Eq. 8).
- **Fermi Energy (E_F):** In the HDR, χ_{xy} is constant (plateau). In the LDR, it depends on $\sqrt{m^2 \alpha^2 + 2m E_F}$.
- **Chirality:** The contribution comes from the difference in transport times and Fermi momenta of the two chiral bands ($\bar{\tau}^+ k_F^+ - \bar{\tau}^- k_F^-$).
- **Fermi Velocity:** Implicit in the group velocity $\mathbf{v}_{\nu}(\mathbf{k})$ and the density of states.

3. Anisotropic Rashba Model (C_{2v} Symmetry)

If the system exhibits anisotropy in effective mass ($m_x \neq m_y$) or Rashba parameters ($\alpha_x \neq \alpha_y$), the Hamiltonian becomes [1]:

$$\hat{H} = \frac{\hbar^2 k_x^2}{2m_x} + \frac{\hbar^2 k_y^2}{2m_y} + \alpha_y k_y \hat{\sigma}_x - \alpha_x k_x \hat{\sigma}_y \quad (11)$$

The Edelstein susceptibility in the HDR depends on the anisotropy ratios $r_m = m_y/m_x$ and $r_{\alpha} = \alpha_y/\alpha_x$:

$$\frac{\chi_{xy}}{\chi_0}(r_m) = \frac{4\pi m_x \alpha r_m}{1 + \sqrt{r_m}} \quad (12a)$$

$$\frac{\chi_{xy}}{\chi_0}(r_{\alpha}) = \frac{4\pi m \alpha_x r_{\alpha}}{1 + r_{\alpha}} \quad (12b)$$

The susceptibility can be boosted by rendering r_m and r_{α} greater than one.

4. Explicit Graphics Description

Based on the extracted data, the following graphics should be generated to visualize the model results:

1. Edelstein Susceptibility vs. Chemical Potential (μ):

- **Description:** Plot χ_{xy}/χ_0 on the y-axis and μ on the x-axis.
- **Behavior:** Shows a plateau in the High-Density Regime (constant value) and a rising curve in the Low-Density Regime.
- **Source:** Figure 2 (Left panel) and Figure 3 (Left panel) in Source [1].

2. Edelstein Susceptibility vs. Rashba Parameter (α):

- **Description:** Plot χ_{xy}/χ_0 on the y-axis and α on the x-axis at fixed μ .
- **Behavior:** Linear increase with α in the high-density regime.
- **Source:** Figure 3 (Right panel) in Source [1].

3. Fermi Surface and Spin Texture:

- **Description:** 2D plot in k_x - k_y space showing the Fermi circles (inner and outer) and the spin orientation vectors (tangential to the circles).
- **Behavior:** Shows spin-momentum locking.
- **Source:** Figure 1 and Figure 2 (Right panel) in Source [1].

4. Anisotropy Dependence:

- **Description:** Surface plot or contour plot of χ_{xy}/χ_0 as a function of r_m and r_α .
- **Behavior:** Susceptibility increases with both ratios, saturating for large r_α .
- **Source:** Figure 6 in Source [1].

5. Scientific Citations

• Rashba Hamiltonian and Basic Formalism:

- Equation (1) and Introduction: Gaillardoni, I., et al., "Edelstein Effect in Isotropic and Anisotropic Rashba Models", *arXiv:2503.20712v1* (2025) [1].
- Original Prediction: Edelstein, V. M., "Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems", *Solid State Communications* **73**, 233 (1990) [29].

• Analytical Expressions (HDR/LDR):

- Equations (8), (9), (10): Gaillardoni et al. (2025) [1].
- Derivations: Appendix A in Gaillardoni et al. (2025) [1].

• Anisotropic Case:

- Equations (11), (12): Gaillardoni et al. (2025) [1].
- Derivations: Appendix B in Gaillardoni et al. (2025) [1].

• Boltzmann Transport Approach:

- Equation (2): Gaillardoni et al. (2025) [1].
- Reference: Inoue, J.-i., et al., "Diffuse transport and spin accumulation in a rashba two-dimensional electron gas", *Physical Review B* **67**, 033104 (2003) [19].

References:

- [1] I. Gaillardoni, M. Trama, A. Maiellaro, C. Guarcello, F. Romeo, and R. Citro, "Edelstein Effect in Isotropic and Anisotropic Rashba Models", *arXiv:2503.20712v1* [cond-mat.mes-hall] (2025).
- [29] V. M. Edelstein, "Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems", *Solid State Communications* **73**, 233 (1990).
- [19] J.-i. Inoue, G. E. Bauer, and L. W. Molenkamp, "Diffuse transport and spin accumulation in a rashba two-dimensional electron gas", *Physical Review B* **67**, 033104 (2003).